CEE498 MLC Project Description | Spring 2026

Important Dates:

Proposal due: April 1st at noon
Progress report due: April 15th at noon
Oral Presentations: April 30 and May 5 (class time)
Final report due: May 11 at 11:59pm

1. Format of the Project

Projects should be done in groups of three. It is important that all team members contribute equally. Acceptable project formats include, but are not limited to:

(a) Implementation or Demo: You can select a research paper related to the topics covered in class and implement their method on a civil engineering dataset. You may want to follow the spirit of recent reproducibility challenges (NeurIPS 2019, ICLR 2019).

(b) Applying the methods covered in this class to your research: You can apply the methodology you learnt in this course to your research. For this to qualify as your class project, you should not have worked on this specific aspect before January of 2026.

2. Project Deliverables (Submissions on Canvas)

2.1. Proposal: The proposal should be uploaded to Canvas in PDF format by one group member. The proposal should include:

- Names of all group members
- A description of the proposed project in 1-2 pages
- Description of the dataset you will be using to train your models
- Expected outcomes (e.g., hypotheses, evaluation metrics, or benchmarks)
- Statement of individual responsibilities and contributions
- Key references, including links to any resources (especially the code and datasets).

You may receive feedback on your proposal (if needed) but not a formal grade. However, failure to turn in the proposal on time will result in a 10% penalty on the project grade.

2.2. Progress update: The progress report should be uploaded to Canvas in PDF format by one group member. The target length is 1-2 pages. It should include:

- Summary of progress to date
- Description of any changes to the original project goals
- If you are doing an implementation project, you should show evidence of implementation progress (e.g., successfully running baseline models on the target dataset)
- Be sure to update the statement of individual responsibilities and contributions, if needed.

As with the proposal, you will receive feedback if needed, but not a formal grade, and failure to submit the progress report on time will result in a 10% penalty on the overall project grade.

2.3. Final Deliverables:

(a) Presentation (50% of the project grade): Each team will deliver a 13-minute presentation followed by a 5-minute Q&A session in class. The Q&A session may include questions from your classmates, the instructor, or the TA. Suggested structure for the presentation:

- *Introduction (~2 minutes):* Introduce group members, describe and motivate the problem or the gap, and briefly present existing solutions if any
- *Methodology (~4 minutes):* Explain your approach, including key ideas, models, or algorithms
- *Results (~5 minutes):* Present experimental setup, evaluation metrics, and key findings
- *Discussion and Conclusions (~2 minutes):* Summarize insights, key takeaways, advantages and limitations, and possible improvements in future works

Formatting Guidelines:

- Slides should be clear, readable, and well-organized, with slide numbers included
- Use figures, diagrams, and visualizations where appropriate instead of dense text
- Avoid overcrowded slides; aim for concise key points
- All plots and figures must have labeled axes and captions where needed
- Text should be large enough to be readable from the back of the room
- Cite any external images, datasets, or materials used
- Avoid large blocks of code; summarize key ideas instead

(b) Final Report (50% of the project grade): The final report must be submitted in PDF format by one group member on Canvas. It should be (the equivalent of) at least six pages (single-spaced, 11-point font, 1-inch margins) and mimic the style of a research paper. Suggested structure for the final report:

- *Title, Authors, and Abstract:* In the Abstract, briefly summarize your problem, line of attack, and most interesting/surprising findings.
- *Introduction:* Define and motivate the problem, discuss background material or related work, and briefly summarize your approach.
- *Methodology/ Approach:* Clearly describe your methods. Include equations, pseudocode, diagrams, or other materials necessary to explain your system. If possible, illustrate intermediate stages with results.
- *Results:* Describe your experimental setup, including datasets, evaluation metrics, and any external code used. Present your quantitative results (if applicable) and show some example outputs. If you are working with videos, provide example outputs via YouTube or another external repository and include links in your report.
- *Discussion and Conclusions:* Summarize key insights from your analysis and experiments. Note that you can still get a good project grade even with mostly negative results, if you show evidence of extensive exploration, thoughtfully analyze the causes of your negative results, and discuss potential solutions.
- *Statement of individual contributions:* Describe the contributions of each member.
- *Statement on use of LLM models:* You are allowed to use LLMs, but not in a way that they replace the required efforts that you are expected to put in. You should include a statement that honestly and thoroughly explains how LLMs were used during the project period.
- *References:* Include all sources, including URLs for datasets, code, and any external materials used.

3. Grading

You must submit your source code for documentation, either as a zipped folder upload or a GitHub repository link. Grading will be based primarily on the quality of your presentation and final report, including the strength of the idea, clarity, thoroughness, depth of analysis, and overall effort. We will run the submitted codes for verification and will deduct points if the code doesn't run properly or lacks sufficient documentation/files.